

Data Fundamentals & Architecture

AWS Certified Machine Learning Associate – Part 2

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Part 1: Types & Characteristics of Data

Understanding data structures and the 3 V's that drive AWS Machine Learning
pipelines

1. Categorizing Types of Data



Structured Data

Highly organized in tabular formats with predefined schemas (rows & columns).

Examples: SQL relational databases, CSV spreadsheets, financial logs.



Semi-Structured

Contains organizational markers/tags without a rigid tabular database schema.

Examples: JSON payloads, XML files, NoSQL document stores (DynamoDB).



Unstructured Data

Lacks predefined conceptual models; represents over 80% of enterprise data.

Examples: Raw images, audio files, video clips, free-form PDF documents.

2. Core Properties of Data (The 3 V's)



Volume

The scale and quantity of generated data, ranging from Gigabytes to Exabytes, requiring elastic storage like Amazon S3.



Velocity

The speed at which data is produced and processed—batch processing vs. real-time streaming (Amazon Kinesis / MSK).



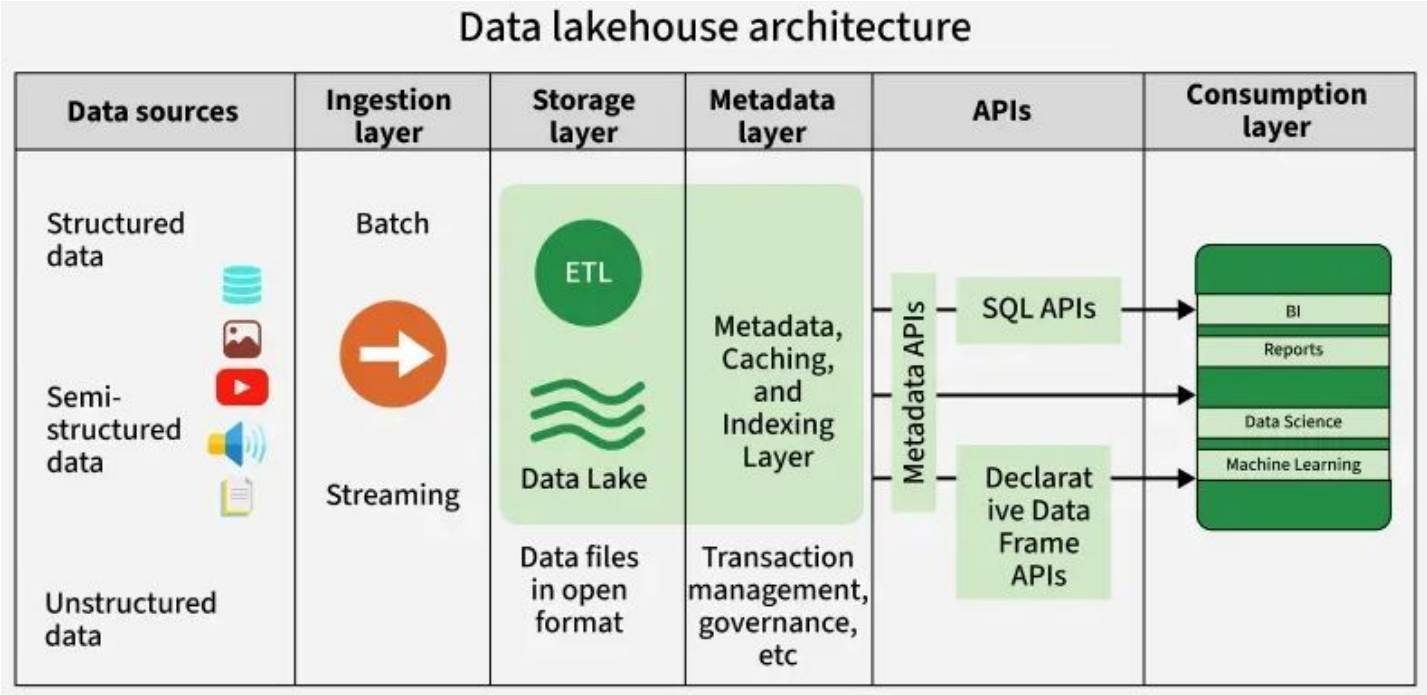
Variety

The diverse range of data types (structured, semi-structured, unstructured) ingested into ML pipelines.

3. Data Warehouse vs. Lake vs. Lakehouse

Architecture	Data Types	Schema Paradigm	Primary AWS Service
Data Warehouse (ETL)	Primarily Structured	Schema-on-Write	Amazon Redshift
Data Lake (ELT)	Structured, Semi & Unstructured	Schema-on-Read	Amazon S3 + AWS Lake Formation
Data Lakehouse	Hybrid (All Data Types)	Unified Governance & ACID	Redshift Spectrum + S3 + Iceberg

4. Lakehouse & Data Mesh Architectures



Data Lakehouse

A hybrid data architecture that directly combines the high-performance analytical capabilities of Data Warehouses with the low-cost scalability of Data Lakes.

Data Mesh Concept

Focuses on **decentralized organizational ownership & governance**. Data is managed as a product by domain teams while adhering to global governance standards.

5. Data Connectivity Protocols: JDBC vs. ODBC

Protocol	Platform Dependency	Language Dependency	Key AWS ML Use Cases
JDBC (Java Database Connectivity)	Platform Independent	Language Dependent (Java/JVM)	Java/Scala-based frameworks like Spark on EMR, Glue ETL jobs.
ODBC (Open Database Connectivity)	Platform Dependent	Language Dependent (C/C++ base)	Native OS client drivers connecting BI tools (Tableau, QuickSight) to Redshift/Athena.

6. Common Ingestion Data Sources

 **Raw Logs:** Application server logs, CloudWatch metrics, and audit trails.

 **REST APIs:** Fetching structured or JSON data from microservices and external web apps.

 **Streaming Data:** Real-time event streams from IoT devices and clickstreams via Amazon Kinesis or Apache Kafka.

7. Row & Text-Based Formats (CSV & JSON)



CSV (Comma-Separated Values)

Plain text format separating records by lines and fields by commas.

Pros: Human-readable, broadly supported by Pandas and SageMaker.

Cons: No embedded schema enforcement, slow query parsing at scale.



JSON (JavaScript Object Notation)

Standard data interchange format between web servers and clients.

Pros: Stores hierarchical nested data alongside key-value attributes.

Cons: Verbose metadata increases file size and reading overhead.

8. Binary Row Storage: Apache Avro

Key Features of Avro

Binary Format: Compact, fast row-based binary serialization format.



Stores Both Data & Schema: Schema is embedded in JSON format within the header.



Schema Evolution: Seamlessly handles adding/removing fields over time.

Ecosystem Compatibility

Optimized for **write-heavy real-time streaming architectures**.

Supported Systems:

- Apache Kafka / MSK
- Apache Spark
- Apache Flink
- Hadoop Ecosystem

9. Columnar Analytics Format: Apache Parquet

Why Parquet for ML & Analytics?

-  **Columnar Storage:** Stores data by column rather than row for aggressive compression.
-  **Predicate Pushdown:** Query engines skip non-relevant columns, cutting query times by 90%+.
-  **AWS Cost Savings:** Drastically reduces data scanned in Amazon Athena & Redshift.

Ecosystem & AWS Integration

The gold standard format for high-throughput batch read queries in ML feature stores.

Supported Systems:

- Hadoop & Apache Hive / Impala
- Apache Spark / PySpark
- Amazon Redshift Spectrum

Summary & Key Takeaways

Mastering data types, lakehouse architectures, connectivity drivers (JDBC/ODBC), and formats (Parquet vs. Avro) is crucial for building cost-effective AWS Machine Learning pipelines.